MENISCUS SHIELD

PHYSICAL HEALTH PRODUCT

Human-Computer Interaction

Wearable Device

Knee injuries and discomfort are common across all ages, affecting everyone from athletes to everyday users. Despite this, many existing solutions are either bulky, uncomfortable, or lack long-term usability.

This project introduces a wearable knee support device that combines ergonomic design, advanced material use, and a clean, future-forward aesthetic. It aims to provide effective joint support, improve user comfort, and promote safer movement in daily life — not only for recovery, but also for prevention.

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INSPIRATION

About Her Story

Last spring, my friend Liu, an experienced hiker, suffered a sudden meniscus flare-up during a hike, leaving her unable to descend and requiring rescue. Doctors later advised six months of rehabilitation, noting the injury could have been avoided with earlier monitoring.

The Missed Signal

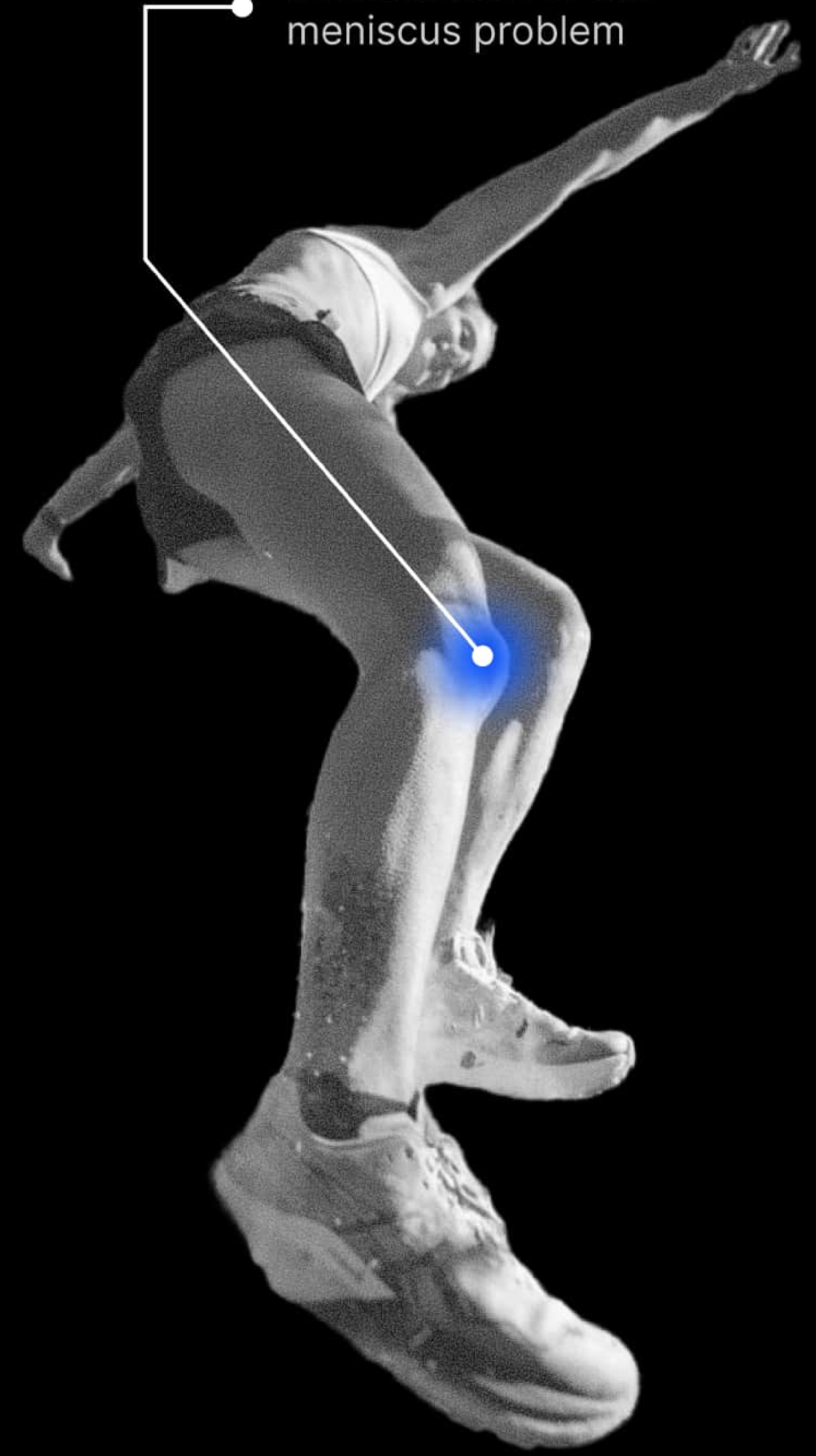
My friend Liu, a seasoned hiker, experienced a sudden worsening of a long-standing meniscus issue while hiking.

Emergency Descent

Eventually, she had to call for rescue. She was later informed that a minimum of six months of rehabilitation would be required.

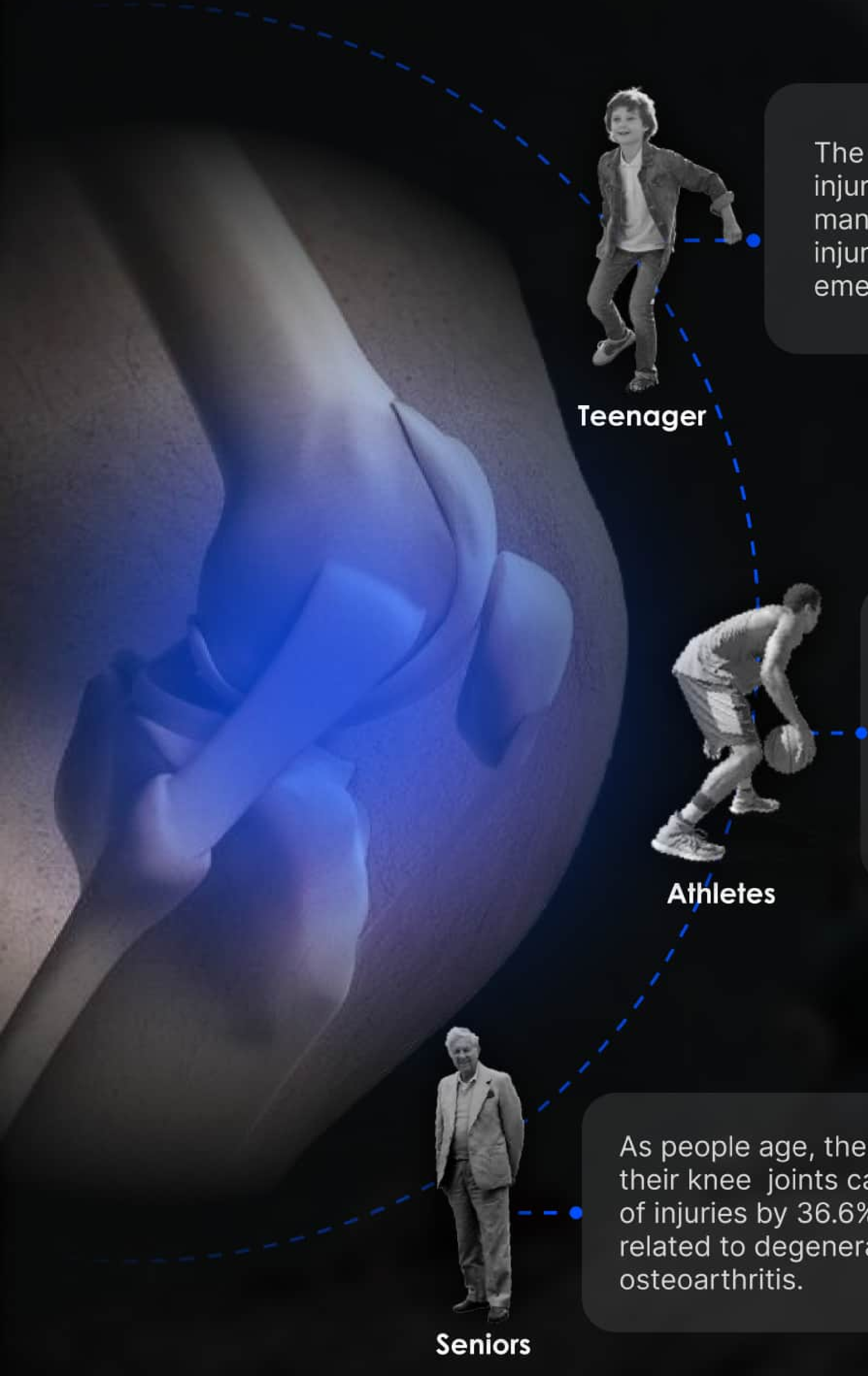
Could've Be Prevented

She had to call for rescue and was told she'd need six months of recovery. The doctor said the injury might have been avoided with earlier monitoring.



RESEARCH

Status Of The Disease



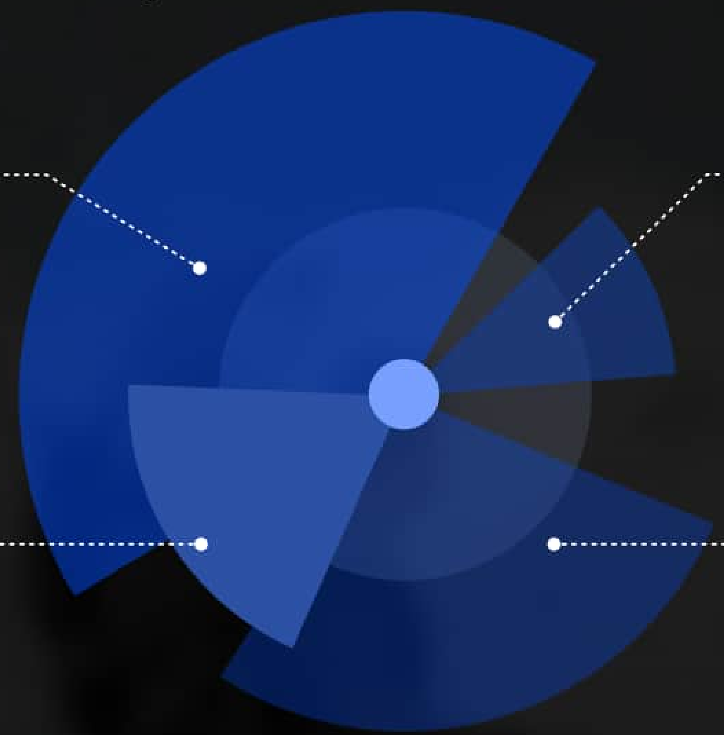
The knee is the most commonly injured joint in adolescents, with many of the 2.5 million annual sports injuries in this group presenting to emergency departments.

People who participate in sports that involve a lot of jumping, running, or sudden changes in direction, such as basketball, soccer, football, and skiing, are at a higher risk of knee injuries.

As people age, the wear and tear on their knee joints can increase the risk of injuries by 36.6%, particularly those related to degenerative conditions like osteoarthritis.

Causes Of Meniscus Injuries

Data show clear gender differences: traumatic injuries are more common in males (~3:1), while degenerative injuries are more prevalent in females (~0.71:1).



Sports Injury

About 42% of cases, often seen in sharp stopping rotational movements

Congenital Anomaly

High prevalence of discoid meniscus in Asian populations, about 19%

Obesity, Poor Posture

Obesity or poor posture (e.g., kneeling, cross-legged) affects ~11%.

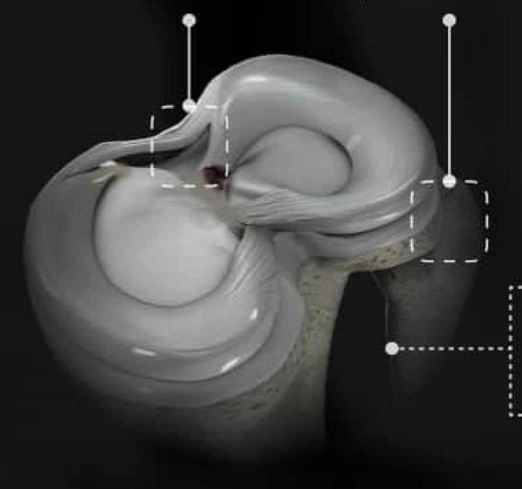
Degenerative Injury

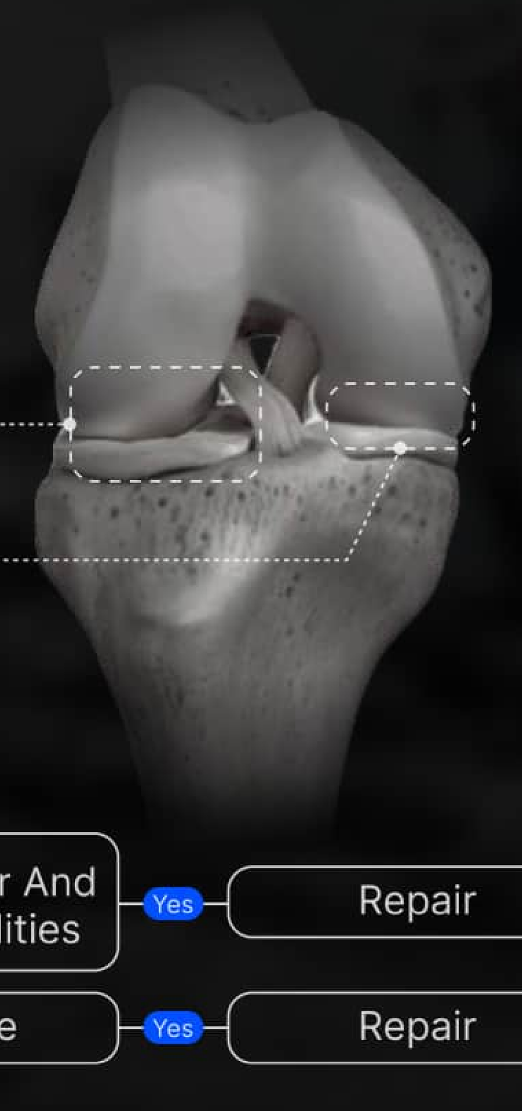
Chronic strain or degeneration affects ~28% of older adults.

Injury & Management

MENISAL TEAR

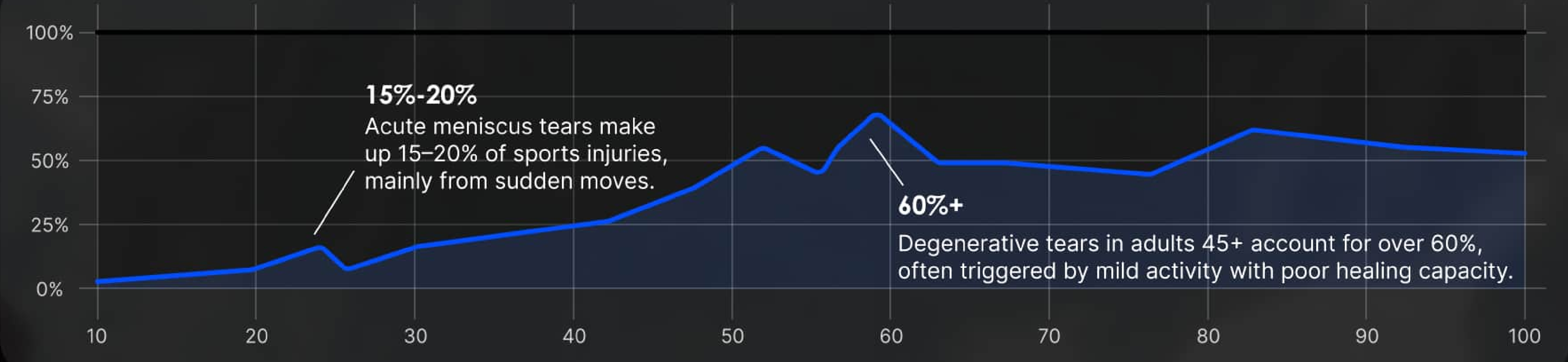
Severe Degenerative Knee
- Yes → Physiotherapy TKR
- No → Symptomatic?
  - No → Displaced?
    - Yes → Consider Arthroscopy
    - No → Physiotherapy And Review In 3 Month
  - Yes → Consider Arthroscopy
    - White-White Zone → Meniscrctomy
    - Red-Red Zone → Reducible? → Yes → Patient<60 Yr And No Comobidities → Yes → Repair
    - Red-White Zone → Patient<40yr? → Yes → Reducible → Yes → Repair





Meniscus Injuries Vary By Population

Degenerative tears dominate in adults over 45 (≥60%), while acute tears are common in athletes (15–20%). Both groups require early intervention due to high risk of progression or career impact.



Treatment Options For Meniscus Injury

	Meniscus Repair	Meniscectomy	Non-Surgical Treatment
Indications	Young Patients; Peripheral Tears	Severe/Central Tears; Elderly	Mild Symptoms; Poor Surgical Fit
Advantages	• Preserves Structure • Better Long-Term Outcome	• Fast Pain Relief • Short Recovery	• No Surgical Trauma Involved • Low Risk Treatment Option
Disadvantages	• Long Recovery Period • High Surgical Demand	• Higher Arthritis Risk • Loss Of Stability	• Long-Term Risk Of Worsening
Risks	10–15% Poor Healing, Adhesions	Infection <1%, Chronic Pain (5–8%)	Atrophy, Delayed Treatment Risks
Outcomes	80–90% Normal Life 60–70% Return To Sport	30–40% Arthritis Risk Rate 90% Pain Relief Short-Term	50–60% Achieve Partial Relief 30–40% Need Surgery Later

ANALYSE

Persona

Zhiyuan

62 | Retired Architect

Urban-Fringe Area, China

Post-Operative Meniscus Injury (Left Knee)

Hiking · Mountain Landscapes · Rehabilitation Tech · Ineffective Devices · Ergonomics · Surgery Risks

Like

Chen enjoys hiking and staying connected to nature, especially mountainous landscapes.He values smart rehabilitation tools that align with his interest in ergonomics.

Technical Acceptance

Mountaineering Hobby

Dislike

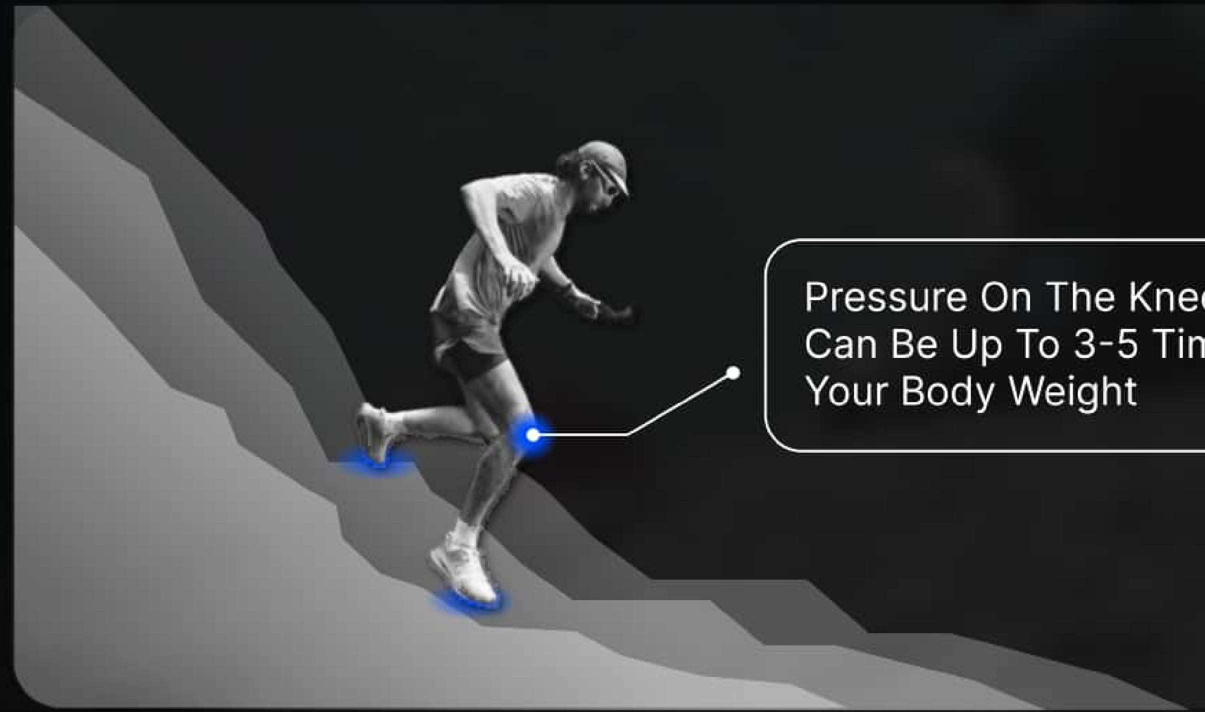
He dislikes bulky or rigid knee braces that restrict movement or feel unnatural.Generic advice like "just avoid hiking" frustrates him due to its lack of nuance.

Pain Point

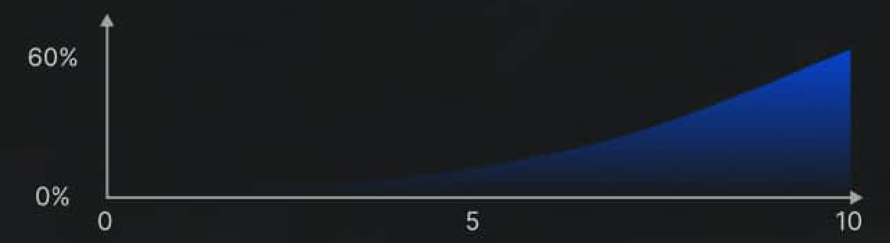
Lingering knee pain limits his confidence and keeps him from fully enjoying outdoor activities. He worries about re-injury and long-term joint degeneration despite his recovery efforts.

"I Don't Want To Be Just A Patient—I Want To Hike Again, Just More Intelligently This Time."

The Effect Of Hills On The Meniscus



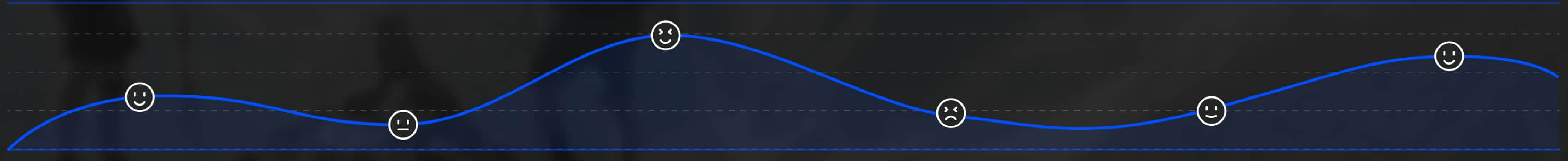
Pressure On The Knee Can Be Up To 3-5 Times Your Body Weight



Older hikers face higher meniscus wear risk.

Hikers with support gear (1) have 30% lower meniscus injury risk than those without (2).

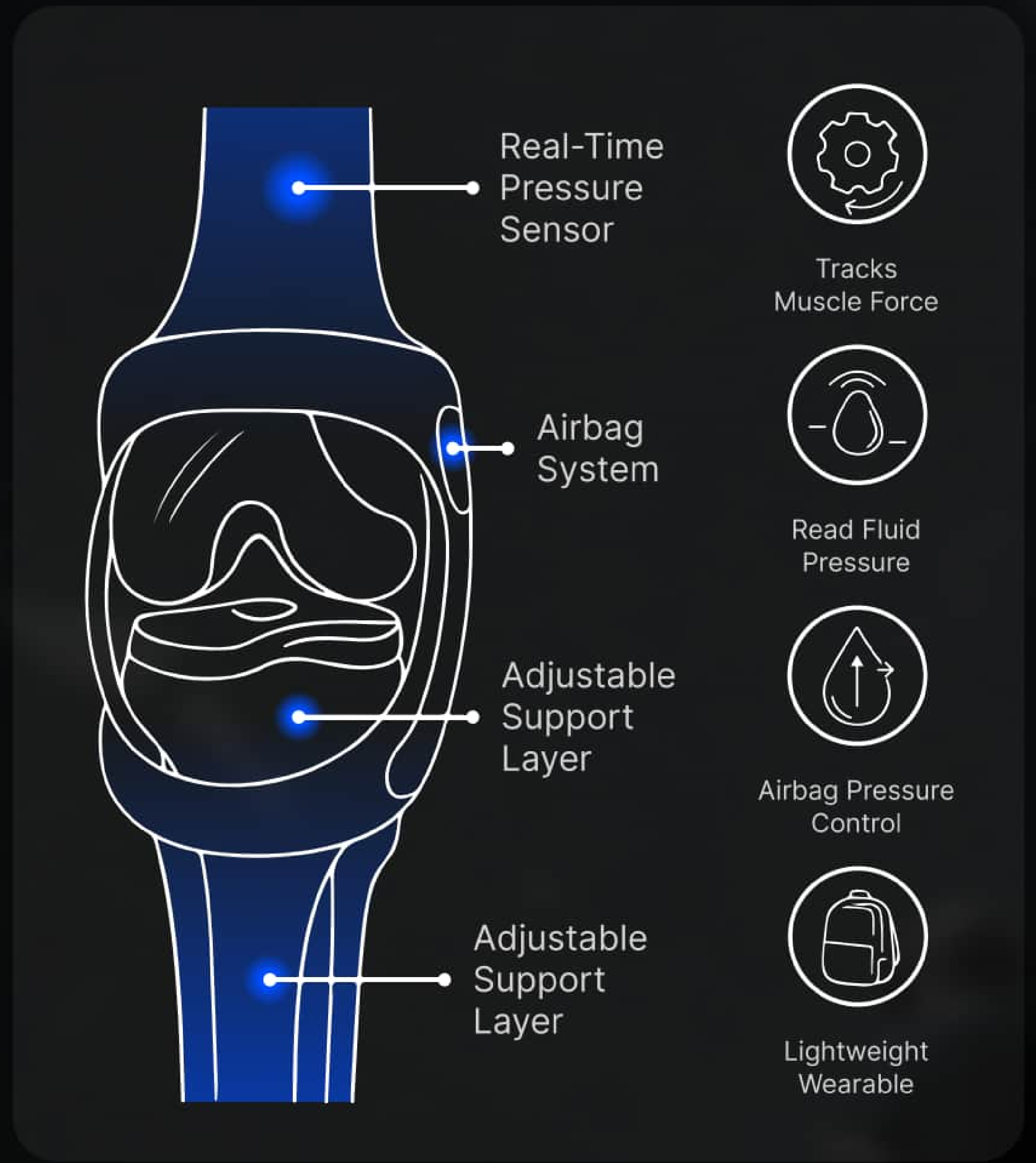
User Journey Map

Stage	Preparation Stage	Start Of The Hike	At The Summit	Descending The Mountain	Drive Home	After Getting Home
Action	Pack the gear, including bringing knee pads. Drive to the mountains and enjoy the scenery on the way.	Started hiking and was excited to take photos along the way. Note the knee bending angle.	Reach the top, sit on the ground, rest and enjoy the feeling of accomplishment.	Start the descent slowly, watching each step to protect your knees from strain or sudden impact.	Driving home, rather tired and half-lost in thought.	Return home for a cold compress and relaxation of the knee.
Experience	Pleased and excited about upcoming outdoor activities.	Feeling the physical exertion while enjoying the view. There was some pain in the knee from bending too much up the hill	Pride in reaching the summit gave way to discomfort—his knees, though briefly at rest, felt stiff and bound when he sat down.	The knee is overloaded on descents and the strength of the knee brace does not allow the knee problem to be relieved	Prolonged use of the knee and driving makes the knee more and more painful	It's warm and cosy, but there's an underlying discomfort that lingers.
Emotion						
Pain Point	Ordinary rehab knee pads don't work well for everyone's requirements	The brace doesn't control the degree of knee flexion very well	Knee pads are still tight when the knee is in a relaxed state and are not suitable	Knee pads do not provide higher protection on the descent	It's too much on the knees at the end of the day.	Ice briefly eases the pain, but doesn't improve the condition.
Opportunities	The strength of knee pads can be more personalized according to the needs of different sports states			At the same time, it meets the need for portability, stability, and freely adjustable sizing.		

DESIGN

Design Goals

Muscle strength and knee joint force vary with different movements. The knee brace should monitor joint cavity fluid pressure in real time and adjust its support accordingly.



Smart Adaptive Knee Brace

To meet diverse user needs, it must balance structural strength with lightweight portability, while maintaining a sense of medical reliability.

The brace integrates a retractable airbag system; by dynamically inflating or deflating the airbags based on real-time knee pressure data, it modulates support strength and ensures adaptive protection during motion.

Design Iteration

Smartwatch + Motion Algorithm 1

Uses: Detects daily activities via gyroscope & accelerometer; app reports knee stress.

Cons: Low accuracy for complex motions; delayed feedback due to wrist placement.

BCI Rehab System 3

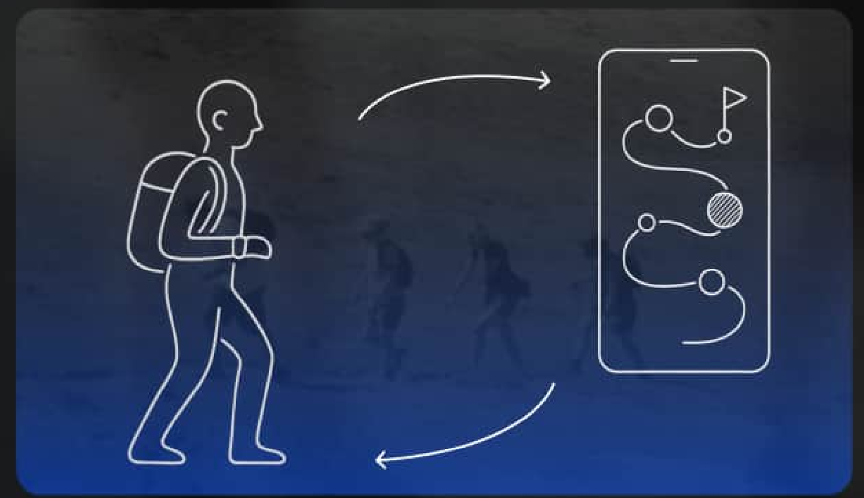
Uses: Predicts motion via brain & muscle signals; gives real-time prompts.

Cons: Expensive and unstable; affected by emotion and fatigue.

Pressure-Sensing Floor + App 2

Uses: Monitors knee load at home; app shows heatmaps & corrections.

Cons: High cost, home renovation required; not portable, unsuitable for outdoor use.

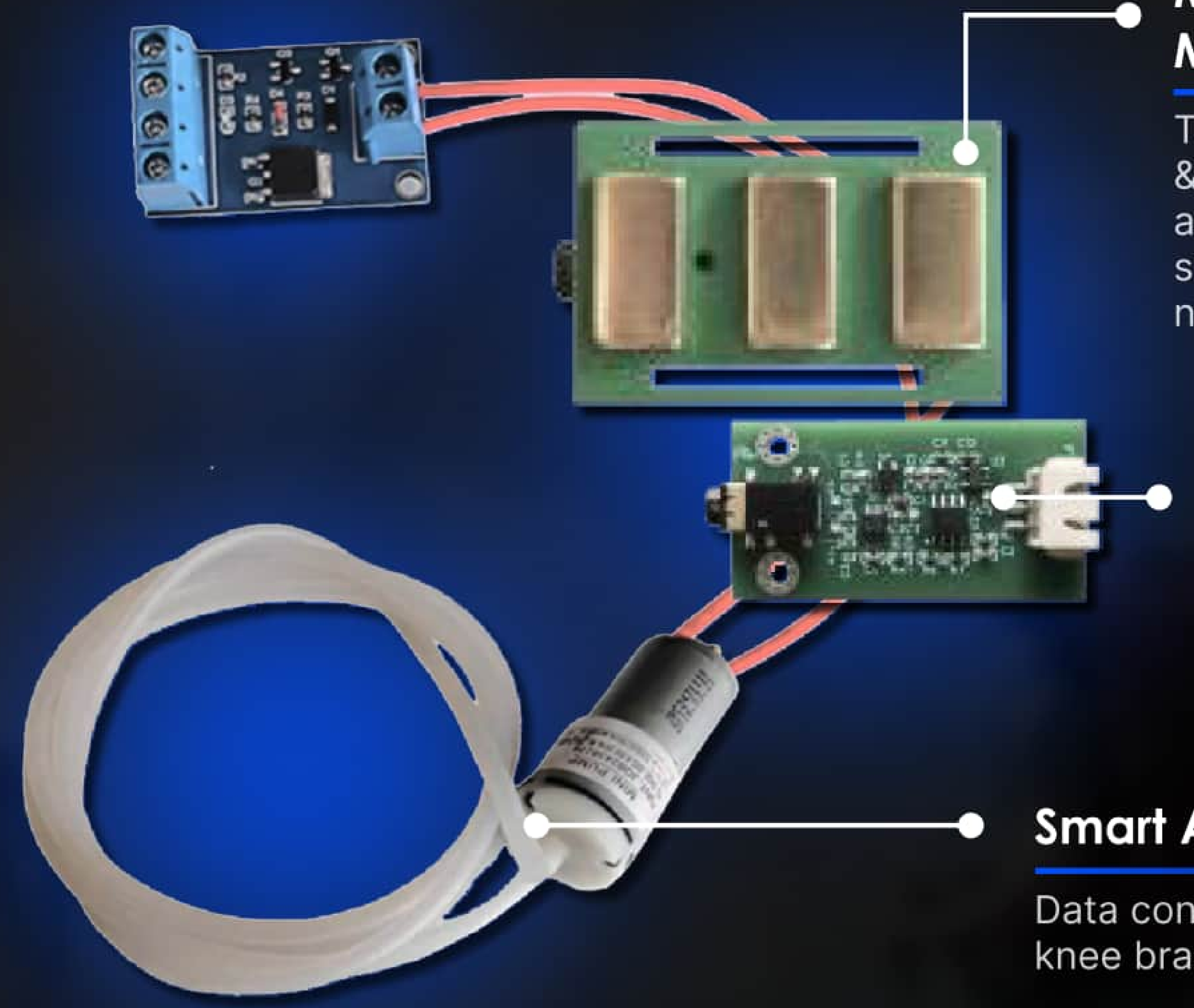


Trail Safety Mapping System 4

Uses: Aggregates sensor data to map terrain risk; app suggests safer routes.

Cons: Niche users; accuracy depends on input quality and large-scale participation.

Design Iteration



MRI and Myoelectricity

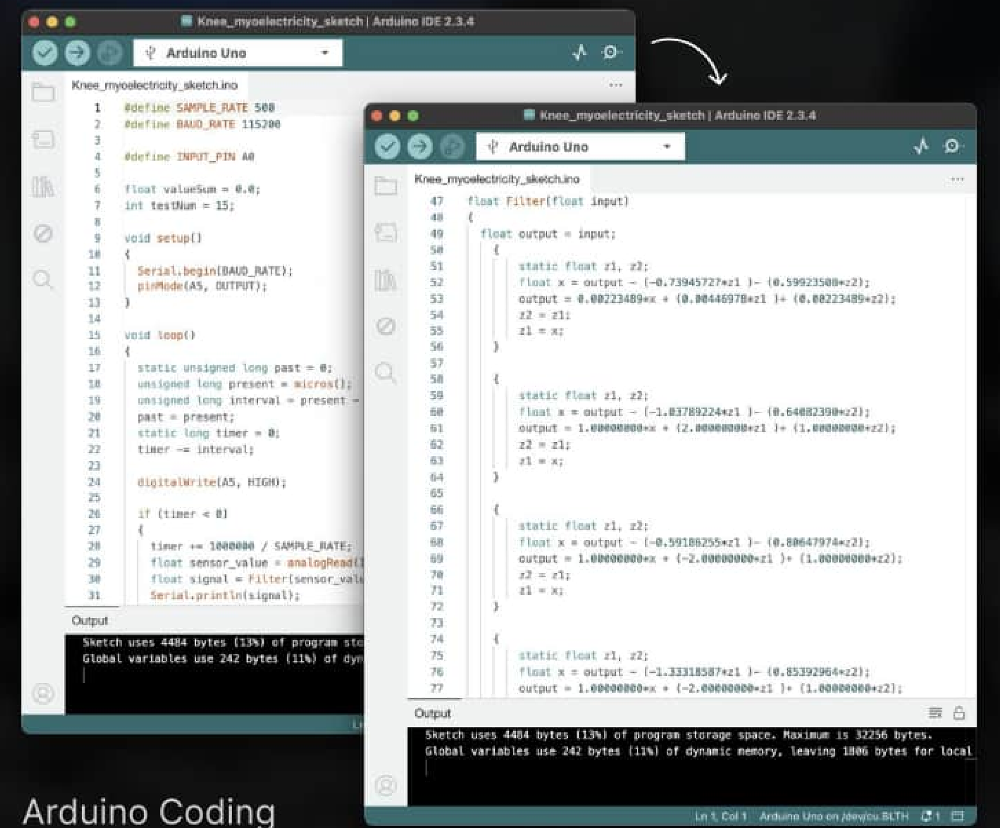
Tracks meniscus via MRI & sEMG Monitors motion and pressure in real time sEMG reflects muscle and nerve activity

Microcontrollers and Signal Processing

Microcontroller processes MRI and sEMG signals to analyze joint condition.

Smart Adjustment

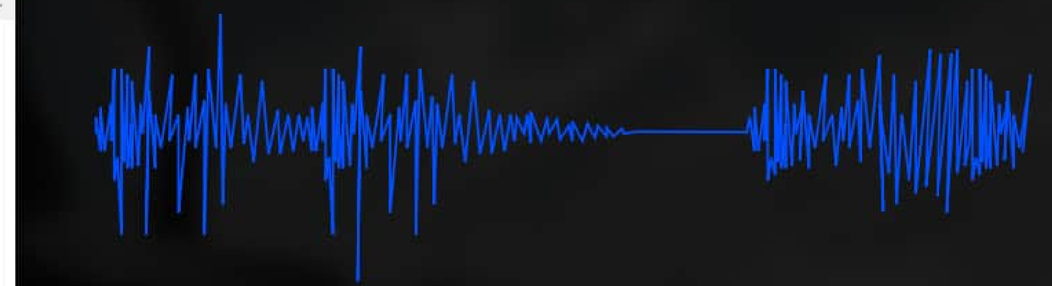
Data controls output module to adjust knee brace pressure via gas flow.



Arduino Coding

Feedback and Monitoring

Curved State:



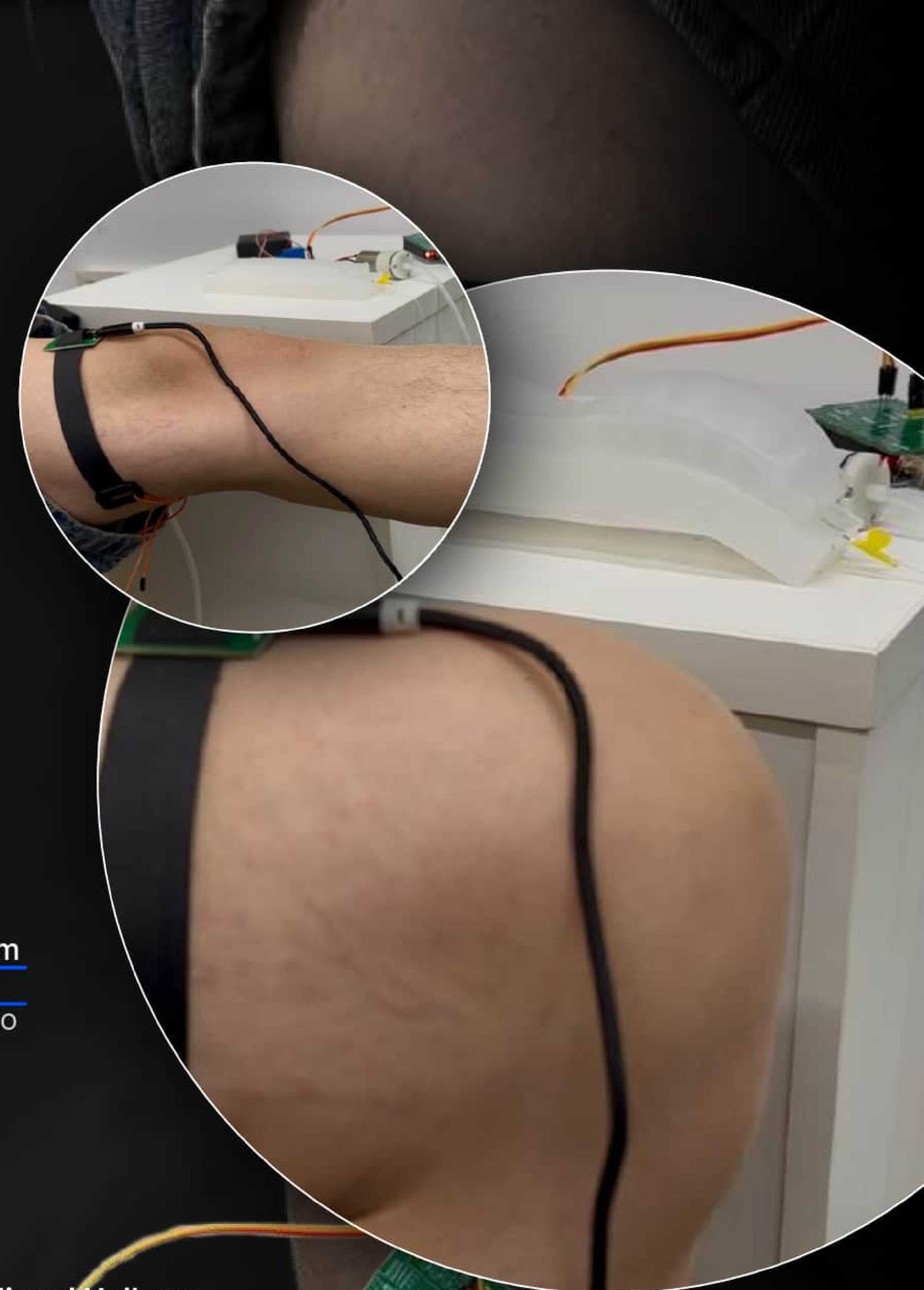
Upright State:

Follow-up continuous monitoring and real-time data is transmitted to a smartphone aor background so that users can monitor their knee status in real time.

Pressure Reset Mode

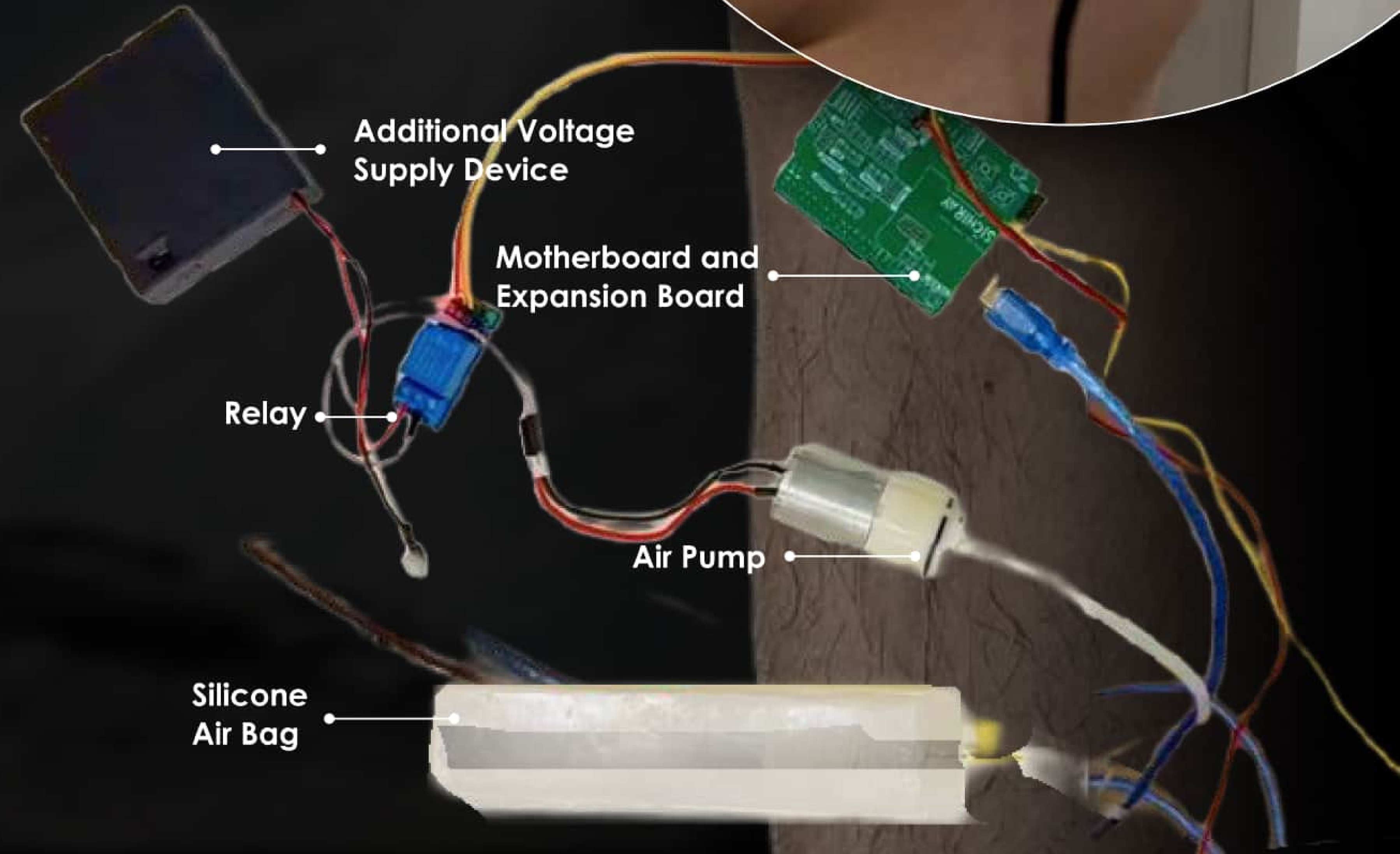
When the leg returns to a straight position, the system detects the change and releases air from the airbag, shifting it into a standby mode.

This helps reduce unnecessary pressure and prepares the system for the next movement cycle.

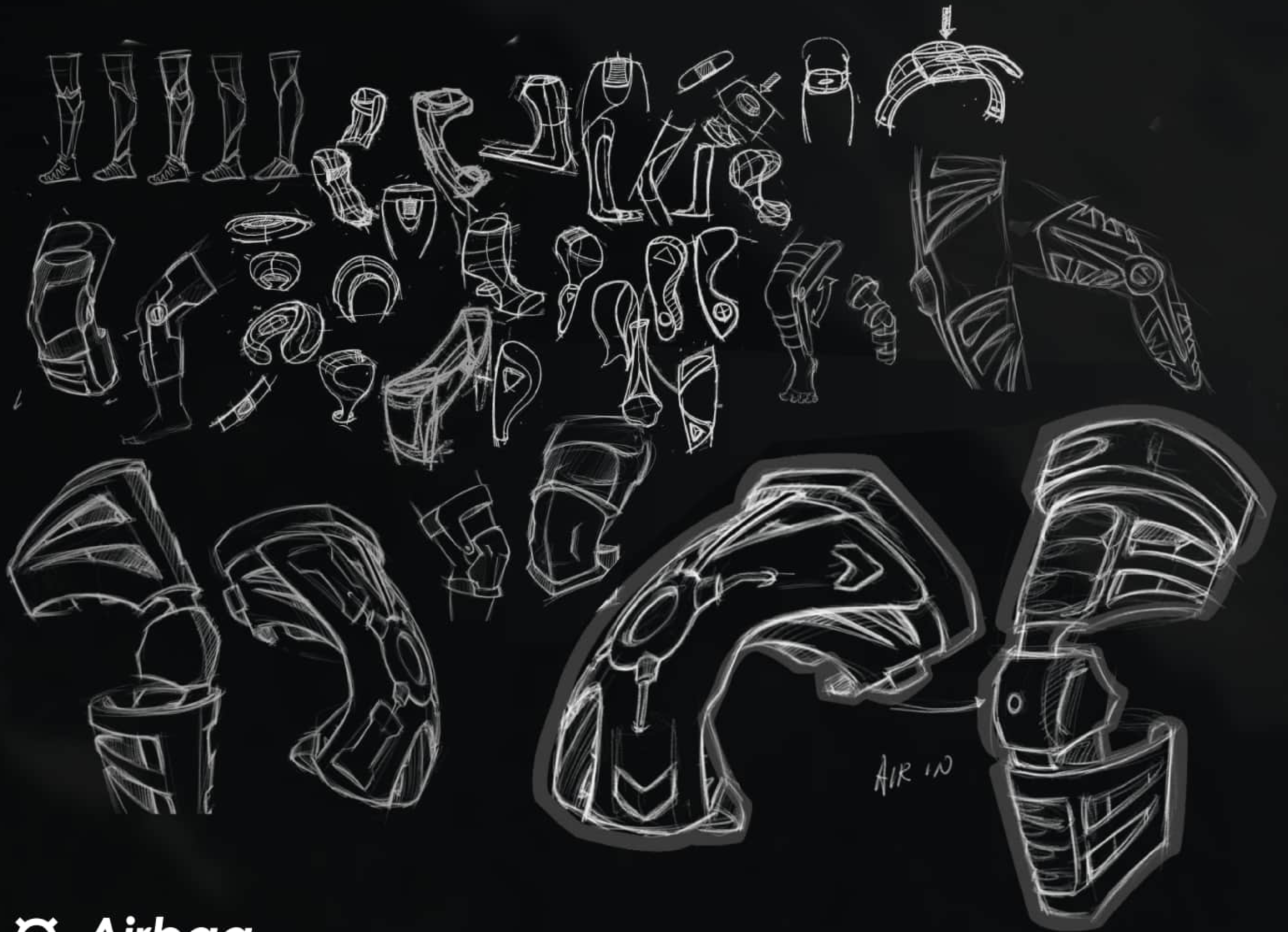


Support Inflation Mode

When the leg bends, the system senses the motion and injects gas into the airbag, causing it to inflate and provide increased support to the knee.



Sketch



Airbag

I use silicone for casting to create an airbag with precisely designed internal cavities, allowing controlled inflation and deflation to support joint movement effectively.

The airbag deflates when the user's knee is fully extended.

Deflate

The aircell activates when the user's knee bends.

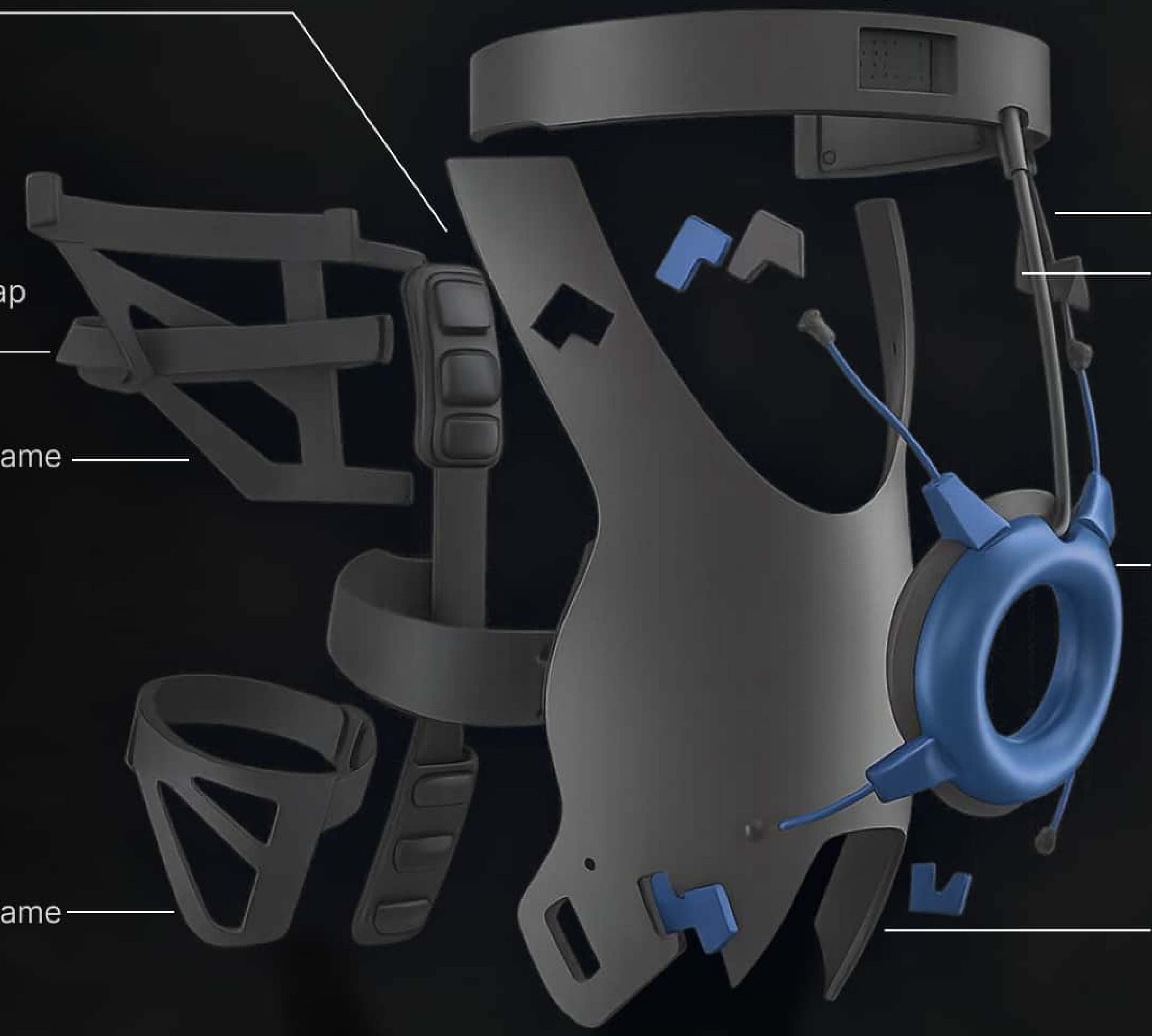
Deflate

Front View

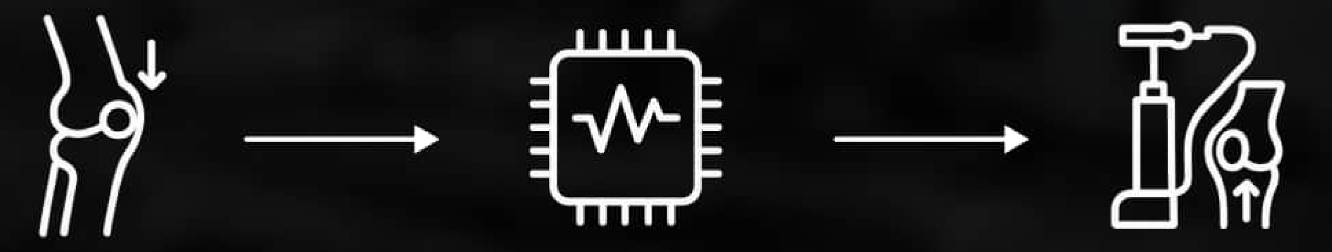
Back View

Top View

Exploded View



Working Process



Sensors monitor pressure levels in the knee joint.

Sensors analyze data and control the air pump.

Adjusts gas volume based on real-time pressure feedback.

This brace features a dynamic airbag system that adapts to the wearer's movements, offering tailored support and comfort. Real-time monitoring ensures optimal knee stability.